

RUDRA PATEL

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EDUCATION

University of Pennsylvania - Philadelphia, PA

BSE in Mechanical Engineering and Applied Mechanics

May. 2030

- **Relevant Extracurriculars:** American Society of Mechanical Engineers (ASME), Penn Aerial Robotics (PennAiR), Penn Jet Propulsion

PROJECTS

Charge Robotics - Marlton, NJ

Sep. 2022 - June. 2026

Executive Board, Robot Operator, GitHub Specialist

- Directed 5+ members to engineer solutions for integrating software with 5+ subsystems
- Created 3 PID velocity and positional controllers with Java and tuned them to smoothly control all subsystems
- Increased performance of 4+ end effectors by maximizing motor and sensor use
- Taught and onboarded software members with GitHub and VS Code; managed team's repositories
- Operated robot's end effectors at competitions and spearheaded strategy sessions; led team to district finalists

Unified Robotics - Marlton, NJ

Sep. 2024 - June. 2026

Student Mentor

- Coached students through the full build cycle of design and iteration while building teamwork and independent problem-solving
- Instructed 4+ peers w/ disabilities on engineering principles through Lego challenges

Residential Light Usage Tracker - Marlton, NJ

Sep. 2025 - June 2026

Capstone Project

- Utilized Onshape to design automated shut-off device; used CAD to make 5+ parts for light-switch mounted device
- Programmed micro servo motor using Arduino Nano; iterated and tested 6 designs before landing on final choice
- Developed door sensor and companion app with team of four to collect light usage data to reduce household energy costs
- Solution monitors occupancy counts inside a room and automatically turn off residential lights when rooms are empty

Aeronautics Club - Marlton, NJ

Sep. 2024- June 2026

Student Mentor

- Prototyped 2 competition gliders out of balsa wood; completed engineering drawings prior to design execution
- Taught 3+ younger members about lift, drag, and thrust; explained basic flight principles

New Jersey Cybersecurity and Communications Integration Cell - Hamilton, NJ

July. 2025 - Sep. 2025

Cybersecurity Intern

- Researched security of aircraft systems, ground systems, and satellites comprehensively at a high-level and presented findings
- Recommended policy changes to director & officials at HQ; co-led introduction of a cybersecurity newsletter with other interns

LEADERSHIP EXPERIENCE

India Temple Association - Berlin, NJ

Aug. 2022 - Aug. 2026

Student Leader (Volunteer Teacher)

- Taught 55+ 6th graders Indian History and Values by creating fun/engaging educational content
- Helped raised \$5000+ for various charities through annual class charity acts; volunteered 350+ hours
- Mentored 2 younger high school student leaders in lesson planning and classroom engagement
- Helped younger student leaders gain confidence to lead classroom sessions independently

Umiya Dham Hindu Temple - Edison, NJ

Sep. 2025 - Feb. 2026

Digital Media Director

- Managed temple Facebook/Instagram; created 220+ recap and news posts; increased followers by 480+
- Promoted temple and raise awareness about festivals; volunteered 60+ hours

WORK EXPERIENCE

Rally House - Mount Laurel, NJ

June 2026 - Aug. 2026

Sales Associate

- Delivered excellent customer service at the point of sale at the register; customer email capture rate was 85%+
- Maintained knowledge of products and arranged merchandise in a visually appearing fashion; kept store clean with teammates
- Created meaningful connections with customers when they entered and left the store

SKILLS

Programming: Java, Python, Visual Basic

Tools & Platforms: Solidworks, Onshape, GitHub, Visual Studio, Microsoft Suite, Eclipse, Visual Studio Code, Claude Code

Languages: English, Gujarati, Italian